Using a filter requires the following syntax – `php://filter/read=<filter>/resource=<file_or_url>` – like so:

```
$text = file_get_contents(
    'php://filter/read=string.toupper/resource=./data/sample.txt'
);
print $text;
```

```
Output:
STAR TREK: ENTERPRISE WAS UNDERRATED.
THE XINDI WERE BETTER ANTAGONISTS THAN THE SULIBAN.
```

Stream Functions

The functions we've seen already in this section (such as `fopen()`, `fgets()`, `file_get_contents()`, and so on) all work on the principle of obtaining a special resource variable that represents a stream. If it hasn't become apparent by now, let's explicitly state what a stream actually is. At its most fundamental, a stream is simply a sequence of bytes. The bytes encode data, which becomes information when that data is handled in the correct context. Later chapters will expand on this idea; for now, just understand that streams are how programs send and receive data. The PHP stream resource, also known as "file handle" (even when that "file" is an abstraction of something else), has mechanisms built into it for operations such as consuming bytes, moving the pointer around, appending bytes, flushing, filtering, and so on. PHP has dozens of stream functions, which are intended to provide interfaces for controlling those stream operations.

I could at this point spend dozens of pages detailing all of the stream functions and no doubt put most readers to sleep or, worse, put them off the book entirely. So let's not do that – at this stage, there's more fundamental things to cover, like numbers, strings, arrays, functions, objects, and much more. However, once those fundamentals are in place, it might be a good idea to take a look at PHP's streams API. One day you might need to write PHP that can take dozens of gigabytes of data, transform it, and serve it to an HTTP client, and using streams becomes necessary to work around memory limitations. Here's a small selection of stream functions and a description of what they do, and you can learn more about them at https://www.php.net/manual/en/ref.stream.php